

WEIRUI PENG

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EDUCATION

University of Michigan

Sep 2025 -

Incoming PhD in Computer Science and Engineering

- Advisor: Dr. Xu Wang and Dr. Eytan Adar

Columbia University

Sep 2022 - Dec 2023

MSc in Electrical Engineering

- Advisor: Dr. Lydia Chilton

The Chinese University of Hong Kong

Sep 2018 - Jun 2022

BEng in Electronic Engineering with First Class Honor

- Advisor: Dr. Thierry Blu

PUBLICATION

1. **Weirui Peng***, Yinuo Yang*, Zheng Zhang, Toby Jia-Jun Li. 2025. [GLITTER: An AI-assisted Platform for Material-Grounded Asynchronous Discussion in Flipped Learning](#). *Accepted to UIST 2025*.
2. Qianou Ma, **Weirui Peng**, Hua Shen, Kenneth Koedinger, and Tongshuang Wu. 2024. [What Should We Engineer in Prompts? Training Humans in Requirement-Driven LLM Use](#). *Accepted to TOCHI 2025*.
3. Zheng Zhang, **Weirui Peng**, Xinyue Chen, Luke Cao, Toby Jia-Jun Li. 2024. [LADICA: A Large Shared Display Interface for Cognitive Assistance in Co-Located Team Collaboration](#). *Accepted to CHI 2025*.
4. Kyrie Zhixuan Zhou, **Weirui Peng**, Yuhan Liu, Rachel F. Adler. 2024. [Exploring the Diversity of Music Experiences for Deaf and Hard of Hearing People](#). *Accepted to CSCW 2025*.
5. **Weirui Peng**, Mengyi Wei, and Kyrie Zhixuan Zhou. 2023. [Enhancing User Engagement in AI Auditing Through Gamification and Storytelling](#). *In CSCW Workshop on Supporting User Engagement in Testing, Auditing, and Contesting AI*.

RESEARCH EXPERIENCE

Research Assistant, SaNDwich Lab

Jan 2024 - Aug 2025

Advised by Prof. Toby Jia-jun Li

University of Notre Dame

- Conducted interviews with users to understand their unique needs in team collaboration tools.
- Designed interactive systems for supporting co-located team collaborations and asynchronous discussion in flip learning.
- Papers accepted to CHI 2025 & UIST2025.

Research Assistant, WiNE Lab

Apr 2024 - May 2025

Advised by Prof. Sherry Tongshuang Wu

Carnegie Mellon University

- Designed and implemented a web-based interactive system that transforms user requirements into effective prompts for large language models (LLMs).
- Conducted in-depth data analysis of user study results to evaluate system usability and effectiveness, deriving key findings for further development.
- Paper accepted to TOCHI 2025.

Research Assistant, CMU HCI Institute*Advised by Prof. Niki Kittur & Prof. Nik Martelaro*

Apr 2024 - Sep 2024

Carnegie Mellon University

- Investigated how designers utilize bio-inspiration to improve design outcomes, focusing on the methods and processes employed in translating natural phenomena into innovative solutions.
- Developed workshop agendas and organized sessions to explore how designers identify and extract key elements from bio-inspiration, facilitating deeper insights into their creative processes.

Research Assistant, Information Experience and Accessibility Lab Sep 2023 - Jan 2024*Advised by Prof. Rachel F. Adler**University of Illinois Urbana-Champaign*

- Conducted qualitative content analysis and text mining on Reddit discussions about music experiences of Deaf and Hard-of-hearing (DHH) people.
- Conducted interviews with DHH people to understand their needs in music appreciation.
- Paper accepted to CSCW 2025.

Research Assistant, Computational Design Lab*Advised by Prof. Lydia Chilton*

Jun 2023 - Dec 2023

Columbia University

- Created an AI-powered interactive system to assist design students in crafting digital art portfolios with p5.js.
- Utilized generative AI to co-create visual, audio, and interactive elements based on artists' visions.

AWARDS & HONORS

- Talent Development Scholarship, Hong Kong, 2022
- Second Runner Up & Xiaomi Vela- Smart City IoT Award, HackUST, 2021
- Bronze Medal at National Level, 7th International Internet+ College Students' Innovation & Entrepreneurship Competition, 2021

SKILLS

- **Programming Languages:** Python, JavaScript, LaTeX, MATLAB, HTML, CSS, SQL
- **Tools and Frameworks:** Flask, MySQL, MongoDB, GCP, GitHub, GitHub Actions, Tensorflow, PyTorch
- **Languages:** Mandarin (native), Cantonese (native), English (fluent)